



Predicting Appointment No- Shows Using Deep Neural Networks



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Problem Formulation



No-shows cause financial loss and scheduling inefficiency.



Objective: predict whether a booked patient will attend their appointment.



Model output: probability $\in [0, 1]$ $\rightarrow$ interpreted as risk of no-show.



Goal metric: maximize AUC-ROC and F1 score.



Dataset Overview

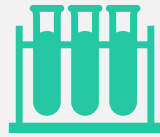
- **Source:** Kaggle – Medical Appointment No-Shows
Rows: 110 527 **Columns:** 14 **Size:** ≈ 4 MB
Target column: **No-show** $\rightarrow$ “Yes” = did not come, “No” = attended
Label ratio: ≈ 20 % no-shows / 80 % shows

Dataset Features

- Patient Demographics
 - **Gender:** Male/Female
 - **Age:** Patient age in years
 - **Neighbourhood:** Patient location (81 unique areas)
- Medical Conditions
 - **Hypertension:** Binary (Yes/No)
 - **Diabetes:** Binary (Yes/No)
 - **Alcoholism:** Binary (Yes/No)
 - **Handicap:** Disability level (0-4)
- Appointment Details
 - **ScheduledDay:** When appointment was scheduled
 - **AppointmentDay:** Actual appointment date
 - **Waiting Time:** Days between scheduling and appointment
 - **SMS_received:** Whether reminder SMS was sent
 - **Scholarship:** Enrollment in welfare program



Planned Data Splits



Training Set: 70% (~77,369 samples)



Validation Set: 15% (~16,579 samples)



Test Set: 15% (~16,579 samples)

• Most Important Predictors

- **Past no-show history:** Strongest predictor across all studies
- **Waiting time:** Days between scheduling and appointment
- **SMS reminders:** Significant impact on attendance
- **Age and medical conditions:** Moderate predictive power
- **Day of week and time:** Appointment scheduling factors



Related Work (Quick Review)

- Multi-Layer Perceptron (MLP) with Attention Mechanism (**Paper:** Ding et al., 2022 - "Machine learning approaches to predicting no-shows in pediatric medical appointment" (Nature - npj Digital Medicine))
- TabNet - Attentive Interpretable Tabular Learning (**Paper:** Arik & Pfister, 2019 - "TabNet: Attentive Interpretable Tabular Learning" (AAAI Conference))
- LSTM/RNN for Sequential Medical Records (**Papers:** Choi et al., 2016 (JAMIA) - "Using RNN models for early detection of heart failure onset")

Multi-Layer Perceptron (MLP) with Attention Mechanism

What they did:

- Built a feed-forward neural network with multiple hidden layers
- Added an attention mechanism that learns which features are most important for each patient
- Handled missing data (77% of their records were incomplete) by using missing indicators as features
- Included weather data as extra features

Results:

- 83% no-show detection rate
- AUC-ROC: 0.85
- False alerts under 17%

TabNet – Attentive Interpretable Tabular Learning



What they did:

- Created a neural network specifically designed for tabular data (not images/text)
- Uses sequential attention – at each decision step, the network picks which features to look at
- Sparse feature selection means not every feature is used for every prediction
- Provides interpretability through attention masks showing which features mattered

Results:

- Outperforms XGBoost and LightGBM on benchmark datasets
- State-of-the-art for tabular classification
- 3-5% AUC improvement over standard MLPs
- Handles high-cardinality categorical features well

LSTM/RNN for Sequential Medical Records

What they did:

- Used Long Short-Term Memory (LSTM) networks to process patient history over time
- LSTM has special gates (input, forget, output) that let it remember long-term patterns
- Processes sequential medical events like multiple appointments, diagnoses, etc.
- Can capture patterns like "patient always misses Monday appointments" or "missed last 3 appointments"

Results:

- ICU mortality prediction: AUC 0.76-0.85
- Beats traditional scoring systems
- Works great when you have patient appointment history

Data Preprocessing & Feature Engineering

Data cleaning:

- Remove invalid ages (**Age** < 0 or unrealistic).
- Convert dates (**ScheduledDay**, **AppointmentDay**) → extract useful features:
 - **waiting_days** = (**AppointmentDay** - **ScheduledDay**)
 - **appointment_weekday**
 - Encoding:
- Convert **Gender**, **Neighbourhood**, **Scholarship**, **SMS_received**, etc. into numeric form.

Normalization:

- Scale continuous features (e.g., **Age**, **waiting_days**) to [0, 1].

Target definition:

- **No-show**: Yes → 1 (did not come)
- **No-show**: No → 0 (came)



Potential Application in Real Systems

- When doctor views upcoming appointments:
 - Display **No-Show Risk: 82% (High)**
 - **Color indicators (Green–Yellow–Red).**
- **Automation examples:**
- High-risk → extra reminder or require deposit.
- Medium-risk → notify doctor or send confirmation request.
- **Business impact:**
- Reduces lost time and revenue.
- Builds trust by improving schedule reliability.